

Qualification Pack



Rooftop Solar Grid Junior Engineer

QP Code: SGJ/Q0106

Version: 4.0

NSQF Level: 5

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SGJ/Q0106: Rooftop Solar Grid Junior Engineer

Brief Job Description

Rooftop Solar Grid Junior Engineer inspects, interconnects and tests different components of the grid connected Solar PV Power Plant in compliance with all relevant codes, standards and safety requirements. To effectively perform the activities, the individual requires a good understanding of design, evaluation, and other technical aspects of solar PV power plant.

Personal Attributes

The individual must be diligent, hardworking, sincere, accountable and should be able to communicate courteously with co-workers and all other stakeholders. He/She should also demonstrate strong work ethics and commitments to complete the job in a time bound manner.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0118: Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant](#)
2. [SGJ/N0119: Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant](#)
3. [SGJ/N0106: Maintain Personal Health & Safety at project site](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Electrical Engineer, General
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2151.0100

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Minimum Educational Qualification & Experience	Completed 4 year UG program (UG Diploma in Electrical/Electronics or equivalent) with NA of experience OR 12th grade Pass (with 1 year of NTC/NAC in relevant trade) OR 10th grade pass (with 3-year Diploma in Electrical/Civil/Mechanical Engineering) OR Previous relevant Qualification of NSQF Level (4.5) OR Previous relevant Qualification of NSQF Level (Solar PV Installer (Suryamitra)) with 3 Years of experience of relevant experience in renewable energy/power sector
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	4.0
Reference code on NQR	QG-05-ES-02627-2024-V2-SCGJ
NQR Version	2

Remarks:

Total 540 hours i.e. 18 Credits (Theory : 150 Hours + Practical : 210 Hours + OJTM : 180 hours)

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SGJ/N0118: Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant

Description

This unit is about performing pre-commissioning activities for solar PV plant interconnections.

Scope

The scope covers the following :

- Introduction to Solar PV Sector in India
- Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant

Elements and Performance Criteria

Introduction to Solar PV Sector in India

To be competent, the user/individual on the job must be able to:

- PC1.** discuss of rooftop solar sector in India.
- PC2.** discuss of Solar PV Technology.
- PC3.** discuss the various types of rooftop Solar PV Power Plants and their working principles
- PC4.** discuss how to identify various system components, their functions and operating principles
- PC5.** discuss key national and state specific policy and regulatory framework on grid interactive PV system
- PC6.** discuss the importance of performing site survey in Roof top Solar PV Power plants
- PC7.** discuss various features of Rooftop Solar Grid connected system with or without battery storage (Hybrid system).
- PC8.** discuss how to identify Instrumentation requirements for inspection and testing of a solar PV power plant
- PC9.** discuss how to acquire skills to work effectively with others while ensuring gender and disability concerns
- PC10.** show how to metering arrangement for grid interactive Rooftop Solar
- PC11.** show how to knowhow of instruments required for inspection and testing of a solar PV power plant
- PC12.** show how to interpret signs, notices and/or cautions at project site

Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant

To be competent, the user/individual on the job must be able to:

- PC13.** discuss how to verify the capacity of the rooftop solar PV power plant as per the relevant policy/ regulation
- PC14.** discuss how to identify the key regulatory parameters for interconnection and metering arrangement including power quality of the grid at the project site
- PC15.** discuss how to identify and verify the documents required for plant interconnection
- PC16.** discuss how to verify that the inverters, panels, protection devices, etc. are conforming to IEC standards or relevant Indian standards

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- PC17.** discuss how to cross verify the components as per the Bill of Quantity (BoQ) and the design Single Line Diagram (SLD)
- PC18.** discuss how to verify and assess the safety of earthing and lightning protection of the plant
- PC19.** discuss how to perform reporting on the inspection findings
- PC20.** discuss how to identify processes where energy/electricity utilization can be optimized
- PC21.** show how to interpret the verification of the components as per the Bill of Quantity (BoQ) and the design Single Line Diagram (SLD)
- PC22.** show how to analyse the PV capacity of the rooftop solar PV power plant as per the relevant policy/ regulation
- PC23.** show how to examine concerned regulations, guidelines & standards for grid interconnection
- PC24.** show how to Pre-commissioning process for grid connected Solar PV System
- PC25.** show how to verify the operation, safety and protections of Grid-Solar PV - Hybrid inverter and battery
- PC26.** show how to examine that the single line diagram of solar PV power plant is as per the regulatory specifications
- PC27.** show how to analyse the checklist for Pre commissioning inspection of Grid connected rooftop solar PV system with or without storage
- PC28.** Illustrate responsible and disciplined behavior's and use of inclusive language that is gender, disability and culturally sensitive

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the keywords and its definitions used in industry
- KU2.** diagnostic/fault finding techniques
- KU3.** various environmental requirements
- KU4.** different types of inverters and protection devices including their relevant regulations for connecting to the grid
- KU5.** types of metering arrangement including "net metering" and "gross metering" concepts and layout
- KU6.** different types of earthing and lightning protection
- KU7.** national technical regulations and technical standards such as Central Electricity Authority's (CEAs) technical standards for connectivity of the Distributed Generation Resources Regulations 2013, CEAs Installation and Operations of Meters Regulations 2007 and CEAs Measures of Safety and Electricity Supply Regulation, 2010
- KU8.** IEEE 1547/IEC 62116 technical standards for utility interconnected photovoltaic inverter for unintentional islanding or any relevant Indian Standards
- KU9.** reporting protocol and documentation required
- KU10.** methods of reducing electrical consumptions
- KU11.** types of meters, bidirectional and unidirectional for measuring import and export of electricity
- KU12.** methods of reducing electrical consumptions
- KU13.** renewable energy sources which can be deployed at the workplace

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KU14. gender and its concepts such as gender roles, gender spectrum, gender as an identity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** timely fill up required project related documentation as per the concerned regulation or guideline
- GS2.** understand manuals, health and safety instructions, regulatory guidelines and other concerned documents
- GS3.** express statements or information clearly so that others can hear and understand
- GS4.** participate in and understand the main points of simple discussions
- GS5.** respond appropriately to any queries
- GS6.** follow organization rule-based decision making process
- GS7.** take decision with systematic course of actions and/or response
- GS8.** follow code of conduct
- GS9.** manage relationships with customers with intent of satisfying the requirements for timely service delivery
- GS10.** recognize problems and search for solutions
- GS11.** choose best methods to complete assigned tasks
- GS12.** approach relevant authority when required
- GS13.** critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS14.** ask questions for better understanding
- GS15.** communicate with team members and colleagues on the significance of greening of jobs
- GS16.** work constructively and collaboratively with others

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar PV Sector in India</i>	27	12	-	-
PC1. discuss of rooftop solar sector in India.	3	-	-	-
PC2. discuss of Solar PV Technology.	3	-	-	-
PC3. discuss the various types of rooftop Solar PV Power Plants and their working principles	3	-	-	-
PC4. discuss how to identify various system components, their functions and operating principles	3	-	-	-
PC5. discuss key national and state specific policy and regulatory framework on grid interactive PV system	3	-	-	-
PC6. discuss the importance of performing site survey in Roof top Solar PV Power plants	3	-	-	-
PC7. discuss various features of Rooftop Solar Grid connected system with or without battery storage (Hybrid system).	3	-	-	-
PC8. discuss how to identify Instrumentation requirements for inspection and testing of a solar PV power plant	3	-	-	-
PC9. discuss how to acquire skills to work effectively with others while ensuring gender and disability concerns	3	-	-	-
PC10. show how to metering arrangement for grid interactive Rooftop Solar	-	4	-	-
PC11. show how to knowhow of instruments required for inspection and testing of a solar PV power plant	-	4	-	-
PC12. show how to interpret signs, notices and/or cautions at project site	-	4	-	-
<i>Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant</i>	24	37	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to verify the capacity of the rooftop solar PV power plant as per the relevant policy/ regulation	3	-	-	-
PC14. discuss how to identify the key regulatory parameters for interconnection and metering arrangement including power quality of the grid at the project site	3	-	-	-
PC15. discuss how to identify and verify the documents required for plant interconnection	3	-	-	-
PC16. discuss how to verify that the inverters, panels, protection devices, etc. are conforming to IEC standards or relevant Indian standards	3	-	-	-
PC17. discuss how to cross verify the components as per the Bill of Quantity (BoQ) and the design Single Line Diagram (SLD)	3	-	-	-
PC18. discuss how to verify and assess the safety of earthing and lightning protection of the plant	3	-	-	-
PC19. discuss how to perform reporting on the inspection findings	3	-	-	-
PC20. discuss how to identify processes where energy/electricity utilization can be optimized	3	-	-	-
PC21. show how to interpret the verification of the components as per the Bill of Quantity (BoQ) and the design Single Line Diagram (SLD)	-	4	-	-
PC22. show how to analyse the PV capacity of the rooftop solar PV power plant as per the relevant policy/ regulation	-	4	-	-
PC23. show how to examine concerned regulations, guidelines & standards for grid interconnection	-	5	-	-
PC24. show how to Pre-commissioning process for grid connected Solar PV System	-	5	-	-
PC25. show how to verify the operation, safety and protections of Grid-Solar PV - Hybrid inverter and battery	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. show how to examine that the single line diagram of solar PV power plant is as per the regulatory specifications	-	5	-	-
PC27. show how to analyse the checklist for Pre commissioning inspection of Grid connected rooftop solar PV system with or without storage	-	4	-	-
PC28. Illustrate responsible and disciplined behavior's and use of inclusive language that is gender, disability and culturally sensitive	-	5	-	-
NOS Total	51	49	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0118
NOS Name	Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Inspection and Testing
NSQF Level	5
Credits	5
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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SGJ/N0119: Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant

Description

This unit is about inspection and commissioning of grid connectivity to solar PV system.

Scope

The scope covers the following :

- Post- commissioning testing of grid interactive rooftop solar PV power system

Elements and Performance Criteria

Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to verify the operation of the installed solar metering system including import and export of energy
- PC2.** discuss and show how to test and verify the inverter operation including anti-islanding functionality, overload, etc.
- PC3.** discuss how to verify the operation of the disconnect protections/isolation devices
- PC4.** discuss how to test, record and verify the power quality of rooftop PV power plant at time of interconnection including harmonics, current, voltage etc.
- PC5.** discuss and show how to test and verify the power factor
- PC6.** discuss how to test and verify the grid connected rooftop solar PV power plant for any phase imbalance
- PC7.** discuss how to test and verify the overall safety of the grid connected rooftop solar PV power plant
- PC8.** discuss how to prepare and complete the relevant documentation
- PC9.** show how to analyse the operation of the installed Solar metering system including import and export of energy
- PC10.** show how to analyse the Test data, power quality of the solar plant
- PC11.** show how to the operation of the disconnect protections/isolation devices.
- PC12.** show how to the know how of test and measure the electrical data

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** reporting protocol and documentation required
- KU2.** company's policies on incentives, personnel management
- KU3.** company's code of conduct
- KU4.** importance of individual's role in the work flow

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- KU5.** organisational culture
- KU6.** effect of harmonic current injection, flicker and DC injection into AC grid on quality of power
- KU7.** inverter response to abnormal conditions like over voltage, under voltage, earth fault, trip function, over and under frequency trip functions and unintentional islanding function
- KU8.** tools & tackles required for inspection and commissioning of the plant
- KU9.** do's and don'ts of DC wiring and installation of other electrical components
- KU10.** testing and commissioning activities and its interpretation - visual inspection, continuity of wiring, earthing, polarity check, insulation, voltage drop etc.
- KU11.** typical faults, their causes and resolution for all system components
- KU12.** connection of the solar power plant to the distribution box/ LT Panel and switchover along with precautions based on current Indian regulations on harmonics, flicker, frequency range and voltage range, etc
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** ways to recognize common electrical problems
- KU15.** common practices of conserving electricity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill in job completion form after inspection and commissioning is completed
- GS2.** read manuals, health and safety instructions, memos, other company's documents
- GS3.** read from different sources- books screens in machines and signage
- GS4.** recognise various color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** express statements or information clearly so that others can hear and understand
- GS6.** participate in and understand the main points of simple discussions
- GS7.** respond appropriately to any queries
- GS8.** follow organization rule-based decision making process
- GS9.** take decision with systematic course of actions and/or response
- GS10.** plan and organize work to meet deadlines
- GS11.** work constructively and collaboratively with others
- GS12.** follow code of conduct
- GS13.** recognize problems and search for solutions
- GS14.** choose best methods to complete assigned tasks
- GS15.** approach relevant authority when required
- GS16.** apply domain knowledge, observations and data to select course of action to perform tasks related to Solar Photovoltaic Systems
- GS17.** critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS18.** ask questions for better understanding
- GS19.** read Standard Operating Practices (SOP) documents
- GS20.** communicate with colleagues on the significance of greening of jobs



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- GS21.** make timely decisions for efficient utilization of resources
- GS22.** complete tasks efficiently and accurately within stipulated time
- GS23.** work with supervisors/team members to carry out work related tasks
- GS24.** work constructively and collaboratively with others
- GS25.** evaluate information obtained from other departments and stakeholders

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant</i>	25	25	-	-
PC1. discuss how to verify the operation of the installed solar metering system including import and export of energy	3	-	-	-
PC2. discuss and show how to test and verify the inverter operation including anti-islanding functionality, overload, etc.	4	5	-	-
PC3. discuss how to verify the operation of the disconnect protections/isolation devices	3	-	-	-
PC4. discuss how to test, record and verify the power quality of rooftop PV power plant at time of interconnection including harmonics, current, voltage etc.	3	-	-	-
PC5. discuss and show how to test and verify the power factor	3	4	-	-
PC6. discuss how to test and verify the grid connected rooftop solar PV power plant for any phase imbalance	3	-	-	-
PC7. discuss how to test and verify the overall safety of the grid connected rooftop solar PV power plant	3	-	-	-
PC8. discuss how to prepare and complete the relevant documentation	3	-	-	-
PC9. show how to analyse the operation of the installed Solar metering system including import and export of energy	-	4	-	-
PC10. show how to analyse the Test data, power quality of the solar plant	-	4	-	-
PC11. show how to the operation of the disconnect protections/isolation devices.	-	4	-	-
PC12. show how to the know how of test and measure the electrical data	-	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0119
NOS Name	Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Inspection and Testing
NSQF Level	5
Credits	4
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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SGJ/N0106: Maintain Personal Health & Safety at project site

Description

This unit is about maintaining health, safety and hygiene at workplace.

Scope

The scope covers the following :

- Adopt safe practices at workplace
- Follow emergencies, rescue and first aid procedures
- Follow good housekeeping practices and infection control guidelines

Elements and Performance Criteria

Adopt safe practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area
- PC2.** identify and report any hazards, risks or breaches in site safety to the appropriate authority
- PC3.** follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk
- PC4.** use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition
- PC5.** follow safe practices when working at height and in confined space
- PC6.** handle all required tools, tackles, materials and equipment safely
- PC7.** identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures
- PC8.** apply ergonomic principles wherever required
- PC9.** use safety signs, labels, charts and notices at workplace
- PC10.** identify work safety procedures and instructions for handling heavy components

Follow emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow emergency and evacuation procedures in case of accidents, fires and natural calamities
- PC12.** use appropriate fire extinguishers for different types of fire
- PC13.** administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.
- PC14.** use correct method to move injured person during an emergency

Follow good housekeeping practices and infection control guidelines

To be competent, the user/individual on the job must be able to:

- PC15.** follow recommended personal hygiene, workplace hygiene and sanitation practices
- PC16.** clean and disinfect all material, tools and supplies before and after use

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- PC17.** report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues
- PC18.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of safety drills
- KU2.** importance of working in clean and safe environment
- KU3.** health and safety roles and responsibilities of relevant personnel within and outside the organization
- KU4.** reporting procedures in case of breaches or hazards for site safety, accidents and emergency situations
- KU5.** basic ergonomic principle
- KU6.** key internal and external source of health and safety information
- KU7.** meaning of hazards, risk and near miss
- KU8.** importance of Personal Protective Equipment required for specific job
- KU9.** forms and classification of hazardous substances
- KU10.** health effect associated with exposure to environmental pollution
- KU11.** housekeeping activities relevant to task
- KU12.** symptoms of infection like fever, cough, swelling and inflammation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** fill safety formats for near miss, unsafe condition
- GS4.** identify potential safety risk and report to appropriate authority
- GS5.** communicate and collaborate with others to incorporate sustainable practices
- GS6.** interpret general safety guidelines, labels, charts and signage

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt safe practices at workplace</i>	13	19	-	-
PC1. explain the requirements for safe work area	2	-	-	-
PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority	2	3	-	-
PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk	1	2	-	-
PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition	2	4	-	-
PC5. follow safe practices when working at height and in confined space	1	1	-	-
PC6. handle all required tools, tackles, materials and equipment safely	1	2	-	-
PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures	1	2	-	-
PC8. apply ergonomic principles wherever required	1	2	-	-
PC9. use safety signs, labels, charts and notices at workplace	1	1	-	-
PC10. identify work safety procedures and instructions for handling heavy components	1	2	-	-
<i>Follow emergencies, rescue and first aid procedures</i>	4	4	-	-
PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities	1	1	-	-
PC12. use appropriate fire extinguishers for different types of fire	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.	1	1	-	-
PC14. use correct method to move injured person during an emergency	1	1	-	-
<i>Follow good housekeeping practices and infection control guidelines</i>	4	6	-	-
PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices	1	1	-	-
PC16. clean and disinfect all material, tools and supplies before and after use	1	1	-	-
PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues	1	2	-	-
PC18. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	21	29	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0106
NOS Name	Maintain Personal Health & Safety at project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety , Installation, Solar Panel Installation Technician
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0118.Pre-Commissioning Inspection of the Grid Connected Rooftop Solar PV Power Plant	51	49	0	0	100	40
SGJ/N0119.Post Commissioning Testing of the Grid Connected Rooftop Solar PV Power Plant	25	25	0	0	50	20
SGJ/N0106.Maintain Personal Health & Safety at project site	21	29	-	-	50	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	20
Total	117	133	-	-	250	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.